

By iteration, we have

$$\begin{aligned}
y_i[n] &= y_i[n-2] + \alpha_i[n-1] - \frac{1+(n-3)k}{1+(n-2)k} \alpha_i[n-2] + \alpha_i[n] - \frac{1+(n-2)k}{1+(n-1)k} \alpha_i[n-1] \\
&= y_i[n-2] + \alpha_i[n] + \frac{k}{1+(n-1)k} \alpha_i[n-1] - \frac{1+(n-3)k}{1+(n-2)k} \alpha_i[n-2] \\
&= y_i[n-3] + \alpha_i[n-2] - \frac{1+(n-4)k}{1+(n-3)k} \alpha_i[n-3] + \alpha_i[n] + \frac{k}{1+(n-1)k} \alpha_i[n-1] - \frac{1+(n-3)k}{1+(n-2)k} \alpha_i[n-2] \\
&= y_i[n-3] + \alpha_i[n] + \frac{k}{1+(n-1)k} \alpha_i[n-1] + \frac{k}{1+(n-2)k} \alpha_i[n-2] - \frac{1+(n-4)k}{1+(n-3)k} \alpha_i[n-3] \\
&= \dots \\
&= y_i[1] + \alpha_i[n] + k \sum_{j=2}^{n-1} \frac{1}{1+jk} \alpha_i[j] - \frac{1}{1+k} \alpha_i[1] \\
&= N_{\mathbf{x};v_i} + \alpha_i[1] + \alpha_i[n] + k \sum_{j=2}^{n-1} \frac{1}{1+jk} \alpha_i[j] - \frac{1}{1+k} \alpha_i[1] \quad (\text{Substitute the initial value } y_i[1] = N_{\mathbf{x};v_i} + \alpha_i[1]) \\
&= N_{\mathbf{x};v_i} + \alpha_i[n] + k \sum_{j=2}^{n-1} \frac{1}{1+jk} \alpha_i[j] + \frac{k}{1+k} \alpha_i[1] \\
&= N_{\mathbf{x};v_i} + \alpha_i[n] + k \sum_{j=1}^{n-1} \frac{1}{1+jk} \alpha_i[j]
\end{aligned}$$

Therefore, we conjecture that Equation (6) has a solution

$$\begin{cases} y_i[n] &= N_{\mathbf{x};v_i} + \alpha_i[n] + k \sum_{j=1}^{n-1} \frac{1}{1+jk} \alpha_i[j] \\ y[n] &= N_{\mathbf{x}} + \alpha[n] + k \sum_{j=1}^{n-1} \frac{1}{1+jk} \alpha[j] \end{cases}$$

The following section proves that it is a solution to Equation (6).

B.5 Proof that Equation (7) is a solution to the recurrence Equation (6)

Substitute $n = 1$ into

$$\begin{cases} y_i[n] &= N_{\mathbf{x};v_i} + \alpha_i[n] + k \sum_{j=1}^{n-1} \frac{1}{1+jk} \alpha_i[j] \\ y[n] &= N_{\mathbf{x}} + \alpha[n] + k \sum_{j=1}^{n-1} \frac{1}{1+jk} \alpha[j] \end{cases},$$

we get

$$\begin{cases} y_i[1] &= N_{\mathbf{x};v_i} + \alpha_i[1] \\ y[1] &= N_{\mathbf{x}} + \alpha[1] \end{cases},$$

which satisfies the initial condition

$$\hat{p}_1(v_i|\mathbf{x}) = \frac{y_i[1]}{y[1]} = \frac{N_{\mathbf{x};v_i} + \alpha_i[1]}{N_{\mathbf{x}} + \alpha[1]}.$$

Substitute

$$\begin{cases} y_i[j] &= N_{\mathbf{x};v_i} + \alpha_i[j] + k \sum_{m=1}^{j-1} \frac{1}{1+mk} \alpha_i[m] \\ y[j] &= N_{\mathbf{x}} + \alpha[j] + k \sum_{m=1}^{j-1} \frac{1}{1+mk} \alpha[m] \end{cases}$$

into the right hand side of

$$\hat{p}_n(v_i|\mathbf{x}) = \frac{\frac{N_{\mathbf{x};v_i} + k \sum_{j=1}^{n-1} y_i[j]}{1+(n-1)k} + \alpha_i[n]}{\frac{N_{\mathbf{x}} + k \sum_{j=1}^{n-1} y[j]}{1+(n-1)k} + \alpha[n]},$$

we get

$$\begin{aligned}
\text{numerator} &= \frac{N_{\mathbf{x};v_i} + k \sum_{j=1}^{n-1} (N_{\mathbf{x};v_i} + \alpha_i[j] + k \sum_{m=1}^{j-1} \frac{1}{1+mk} \alpha_i[m])}{1 + (n-1)k} + \alpha_i[n] \\
&= \frac{N_{\mathbf{x};v_i} + (n-1)k N_{\mathbf{x};v_i} + k \sum_{j=1}^{n-1} (\alpha_i[j] + k \sum_{m=1}^{j-1} \frac{1}{1+mk} \alpha_i[m])}{1 + (n-1)k} + \alpha_i[n] \\
&= \frac{(1 + (n-1)k) N_{\mathbf{x};v_i} + k \sum_{j=1}^{n-1} (\alpha_i[j] + k \sum_{m=1}^{j-1} \frac{1}{1+mk} \alpha_i[m])}{1 + (n-1)k} + \alpha_i[n] \\
&= N_{\mathbf{x};v_i} + \alpha_i[n] + k \frac{\sum_{j=1}^{n-1} (\alpha_i[j] + k \sum_{m=1}^{j-1} \frac{1}{1+mk} \alpha_i[m])}{1 + (n-1)k} \\
&= N_{\mathbf{x};v_i} + \alpha_i[n] + k \frac{\sum_{j=1}^{n-1} \alpha_i[j] + \sum_{j=1}^{n-1} (k \sum_{m=1}^{j-1} \frac{1}{1+mk} \alpha_i[m])}{1 + (n-1)k} \\
&= N_{\mathbf{x};v_i} + \alpha_i[n] + k \frac{\sum_{j=1}^{n-1} \alpha_i[j] + k \sum_{j=1}^{n-1} \sum_{m=1}^{j-1} \frac{1}{1+mk} \alpha_i[m]}{1 + (n-1)k} \\
&= N_{\mathbf{x};v_i} + \alpha_i[n] + k \frac{\sum_{j=1}^{n-1} \alpha_i[j] + k \sum_{m=1}^{n-2} \frac{n-1-m}{1+mk} \alpha_i[m]}{1 + (n-1)k} \\
&= N_{\mathbf{x};v_i} + \alpha_i[n] + k \frac{\alpha_i[n-1] + \sum_{j=1}^{n-2} \alpha_i[j] + k \sum_{j=1}^{n-2} \frac{n-1-j}{1+jk} \alpha_i[j]}{1 + (n-1)k} \\
&= N_{\mathbf{x};v_i} + \alpha_i[n] + k \frac{\alpha_i[n-1] + \sum_{j=1}^{n-2} (1 + k \frac{n-1-j}{1+jk}) \alpha_i[j]}{1 + (n-1)k} \\
&= N_{\mathbf{x};v_i} + \alpha_i[n] + k \left(\frac{\alpha_i[n-1]}{1 + (n-1)k} + \frac{\sum_{j=1}^{n-2} \frac{1+jk+kn-k-jk}{1+jk} \alpha_i[j]}{1 + (n-1)k} \right) \\
&= N_{\mathbf{x};v_i} + \alpha_i[n] + k \left(\frac{\alpha_i[n-1]}{1 + (n-1)k} + \frac{\sum_{j=1}^{n-2} \frac{1+(n-1)k}{1+jk} \alpha_i[j]}{1 + (n-1)k} \right) \\
&= N_{\mathbf{x};v_i} + \alpha_i[n] + k \left(\frac{\alpha_i[n-1]}{1 + (n-1)k} + \sum_{j=1}^{n-2} \frac{1}{1+jk} \alpha_i[j] \right) \\
&= N_{\mathbf{x};v_i} + \alpha_i[n] + k \sum_{j=1}^{n-1} \frac{1}{1+jk} \alpha_i[j] \\
&= y_i[n],
\end{aligned}$$

$$\begin{aligned}
denominator &= \frac{N_{\mathbf{x}} + k \sum_{j=1}^{n-1} (N_{\mathbf{x}} + \alpha[j] + k \sum_{m=1}^{j-1} \frac{1}{1+mk} \alpha[m])}{1 + (n-1)k} + \alpha[n] \\
&= N_{\mathbf{x}} + \alpha[n] + k \frac{\sum_{j=1}^{n-1} \alpha[j] + k \sum_{j=1}^{n-1} \sum_{m=1}^{j-1} \frac{1}{1+mk} \alpha[m]}{1 + (n-1)k} \\
&= N_{\mathbf{x}} + \alpha[n] + k \frac{\sum_{j=1}^{n-1} \alpha[j] + k \sum_{m=1}^{n-2} \frac{n-1-m}{1+mk} \alpha[m]}{1 + (n-1)k} \\
&= N_{\mathbf{x}} + \alpha[n] + k \left(\frac{\alpha[n-1]}{1 + (n-1)k} + \frac{\sum_{j=1}^{n-2} \frac{1+(n-1)k}{1+jk} \alpha[j]}{1 + (n-1)k} \right) \\
&= N_{\mathbf{x}} + \alpha[n] + k \sum_{j=1}^{n-1} \frac{1}{1+jk} \alpha[j] \\
&= y[n],
\end{aligned}$$

$$\frac{numerator}{denominator} = \frac{y_i[n]}{y[n]} = \hat{p}_n(v_i|\mathbf{x}).$$

This shows that the obtained solution satisfies the original recurrence equation.

C Experimental details

C.1 Recursive training

We train the LM at generation n exclusively on the text generated by generation $n-1$. In this appendix, we begin by introducing the formal definition of this recursive process. Then, we elaborate on the details, including hyperparameters used in our training.

C.1.1 Formal definition

We define our recursive training process by a first-order Markov process:

$$D \rightarrow \theta[1] \rightarrow \theta[2] \rightarrow \theta[3] \dots,$$

where $\theta[n]$ is the vector of all weights of generation n , subject to the first-order Markov assumption: given the values of all weights of generation n , then the probability of the weights of generation $n+1$ gets a certain set of values, is conditionally independent of the weights of generation $n-1$ and before, including the initial text corpus, D , that is

$$P(\theta[n+1] \mid \theta[n], \theta[n-1], \dots, \theta[1], D) = P(\theta[n+1] \mid \theta[n]).$$

C.1.2 Details and hyperparameters

The transition from $\theta[n]$ to $\theta[n+1]$ is defined by the following stochastic algorithm:

Randomly initializes the weights of generation $n+1$; Update the weights by generative pre-training the LM, that is, gradient descent of the cross entropy loss between the target token and the LM's prediction given a preceding token sequence; The target token is generated by the generation n by faithfully (*i.e.*, with `temperature=1`) sampling from its output distribution given that preceding token sequence. Algorithm 1 is the pseudocode of this algorithm.